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Experimental dataset of sub-critical cylinder wake velocity fields

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Deep learning is of growing interest to the fluids community due its potential applications for real-time prediction and control. Indeed, whereas computational fluid dynamics solvers are prohibitively time-intensive for real-time implementation, deep learning methods can be evaluated in milliseconds once trained. However, the majority of efforts to apply deep learning methods to flows have been performed with computational data rather than experimental data, in part due to the lack of freely available large experimental datasets. In this paper we present a 64.7 GB dataset of two-dimensional velocity fields measured with particle image velocimetry. These fields show the temporal evolution of cylinder wakes in the sub-critical regime for Reynolds numbers ranging from 240 to 2520. The data is broken into discrete time series made up of 100 time-steps each, representing approximately three vortex shedding cycles. The dataset, available for immediate and free use, is poised to facilitate research on deep learning methods in fluids and other endeavors where such a large, detailed experimental dataset is of utility.

Background & Summary/Introduction

Among the most recognizable canonical flows, vortex shedding is famous for its mesmerizing beauty and diverse applications. Indeed, the characteristic pattern of alternating vortices convecting downstream, known as the Kármán vortex street, is observed behind skyscrapers, chimneys, islands, and other bluff bodies submerged in flows. Given the extensiveness of this phenomena, vortex shedding has been studied in detail for decades, giving rise to many comprehensive studies of the features of this phenomenon across Reynolds numbers (Re)[1]. Briefly, around $Re = 50$, an instability occurs in the previously steady recirculation region of the wake of a cylinder, which in turn produces a laminar pattern of vortices. The transition to turbulence begins around $Re = 190$, at which point small-scale three-dimensional instabilities begin to form[2]. Here, we focus on vortex shedding in the "irregular range" or "sub-critical regime" (Re between 300 and 200,000), which is characterized by increasingly prevalent three-dimensional effects, irregular velocity fluctuations, and the transition of the outer shear layer[3] (Fig. 1).

Numerical solvers used for computational fluid dynamics (CFD) are capable of modeling vortical wakes in this regime[4]. While an excellent tool for many engineering problems, these methods are computationally expensive and prohibitively time intensive for real-time applications. As a potential alternative to CFD, data-driven models have received increased attention for various tasks in fluid mechanics, including for predicting, enhancing, and reconstructing fluid flows through machine learning-based models[5–7]. Once trained, these methods benefit from fast model evaluation at low cost, potentially enabling practical real-time, real-world applications. However, data-driven methods require copious amounts of data for training, with most data-driven efforts in fluid mechanics, to date, utilizing numerically-simulated dataset. The reliance on simulated data limits the practicality of these approaches for real-world application, as it removes significant challenges related to observability such as limited and imperfect measurements.

To facilitate the inclusion of experimental data in developing methods for predicting fluid flows and for other efforts where such data would be of use, we present a 64.7 GB dataset of velocity fields in the wake of a circular cylinder, recorded at various Reynolds numbers in the sub-critical regime[8]. We employ digital particle image velocimetry (PIV), a method of quantitative flow visualization which is used here to determine the time-evolving instantaneous flow fields[9]. The data are organized into 100 time-step sequences, with frame rates chosen such that the nondimensional interframe time is approximately constant regardless of Reynolds number. This dataset is available for immediate use at the CaltechDATA repository[8].

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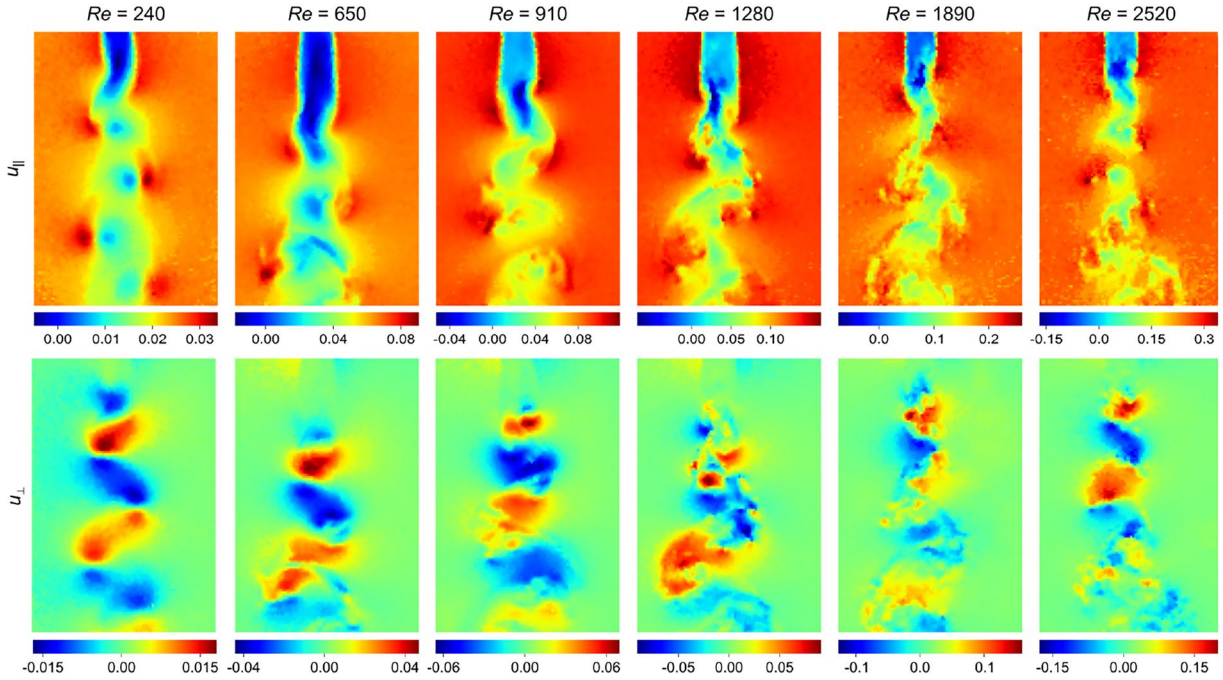


Fig. 1 Streamwise (top) and spanwise (bottom) velocity fields in a cylinder wake for Re between 240 and 2520. Velocity fields taken from accompanying dataset. Colorbar scales are in m s^{-1}.

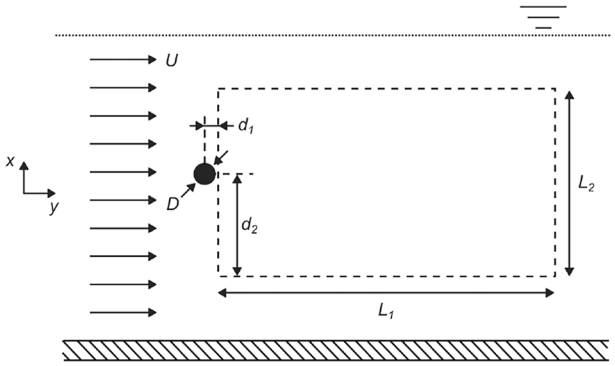


Fig. 2 Schematic of the imaging region of interest, outlined by the black dashed line, relative to the water tunnel and cylinder from the view of the camera. The region of interest was illuminated by a laser sheet, and recorded from the angle shown.

Methods

A small free-surface water tunnel was used in experiments to collect the dataset provided. This tunnel has test section dimensions of 0.15 m (W) × 0.15 m (H) with length 0.61 m. An acrylic cylinder was mounted to span the tunnel walls approximately halfway between the free surface and the tunnel floor (Fig. 2). The diameter of the cylinder used was fixed to $D = 9.53 \times 10^{-3}$ m (cylinder aspect ratio L/D = 15.7), and the inlet velocity was varied between $U = 0.03$ m s^{-1} to $U = 0.26$ m s^{-1} to achieve different Reynolds numbers for the dataset. No free-surface perturbations were observed at the range of Reynolds numbers tested. Flow speeds were determined by taking the mean streamwise flow velocity in a small window in the free-stream. This resulted in Reynolds numbers ranging from 240 to 2520, where we define Reynolds number as

$$\mathrm{Re} = \frac{UD}{\nu}$$

where ν is the kinematic viscosity of water.

The imaging region of interest is shown in Fig. 2, along with the relative geometry of the cylinder and tunnel walls. The dimensions of the region of interest are $L_1 = 0.125$ m ($\approx 13D$) and $L_2 = 0.084$ m ($\approx 9D$). Located at the trailing edge of the cylinder ($d_1 = \frac{1}{2}D$, $d_2 = 0.041$ m), the region of interest was illuminated by a continuous laser sheet. This region was selected to start at the first pixel row not including the cylinder itself, such that we can assume the cylinder center is $0.5D$ upstream of the region edge. A high-speed camera (IDT XSM-3520 set to 2144 × 1440 resolution) was

File name	*Re*	*St*	*U* (m s^{-1})	Frame rate (s^{-1})	*t**	Shedding frequency (s^{-1})	Shedding cycles per sequence
Re_240.h5	242 ± 7	0.16	0.025 ± 0.001	16	0.17	0.42	2.6
Re_650.h5	650 ± 10	0.16	0.068 ± 0.001	40	0.18	1.13	2.8
Re_910.h5	910 ± 14	0.15	0.095 ± 0.001	56	0.18	1.54	2.8
Re_1280.h5	1280 ± 20	0.17	0.135 ± 0.002	80	0.18	2.39	3.0
Re_1890.h5	1890 ± 30	0.19	0.198 ± 0.003	120	0.17	4.03	3.4
Re_2520.h5	2520 ± 60	0.20	0.264 ± 0.006	160	0.17	5.55	3.5

Table 1. Experimental parameters.

File name	Number of sequences	Shedding Cycles	Amount of data (GB)
Re_240.h5	567	1488	10.7
Re_650.h5	934	2638	17.7
Re_910.h5	481	1322	9.1
Re_1280.h5	477	1425	9.0
Re_1890.h5	472	1585	8.9
Re_2520.h5	490	1699	9.3

Table 2. Database size.

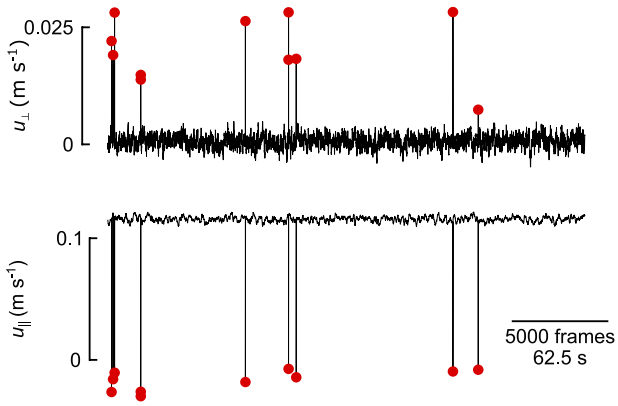


Fig. 3 Example traces of mean spanwise (top) and streamwise (bottom) velocity with recording artefacts identified for one acquisition at $Re = 1300$. Artefacts were identified based on anomalies in the mean spanwise velocity and are indicated on the trace as red circles.

used to record the flow in this region. The frame rate of the camera was adjusted based on the mean flow speed to maintain a near-constant non-dimensional time between frames regardless of tunnel speed. The non-dimensional time, otherwise known as the formation time, is given by Jeon & Gharib (2004)[10] for a cylinder flow as

$$t^* = \frac{Ut}{D}$$

where t^* is the non-dimensional time and t is the dimensional time. Frame rates were selected and the data was subsequently processed such that this value would be approximately 0.18 across all Reynolds numbers tested. The exact non-dimensional inter-frame times, frame rates, Reynolds numbers, shedding frequency, and sequence lengths are available in Table 1.

The velocity fields were calculated from image pairs using PIVview by PIVTEC, a commercial PIV software. An interrogation window of 32 × 32 pixels was used during processing, with 50% overlap. This resulted with a spacing of 9.388×10^{-4} m between processed velocity datapoints in the resulting fields, corresponding to ~0.1D relative to the cylinder diameter.

Data Records

The data is available online on the CaltechDATA repository[8]. Experiments were conducted at six Reynolds numbers (Table 1). Velocity field data from each Reynolds number is stored in a separate Hierarchical Data Format file (h5). Within each file, there are two fields, 'u_x' and 'u_y', corresponding to the streamwise and spanwise components of the velocity field, respectively. The data have been parsed into *n* sequences, each 100 timesteps

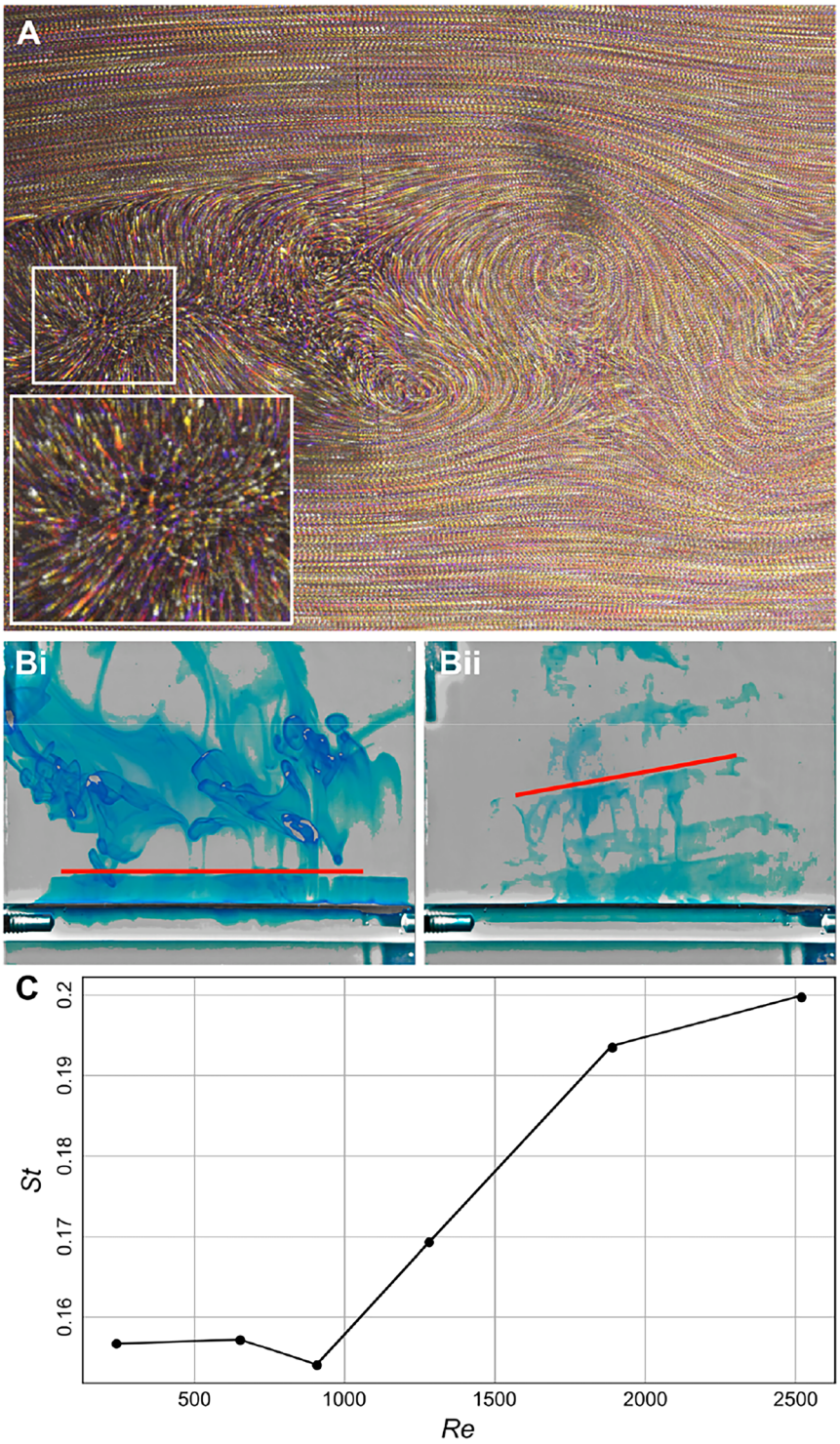


Fig. 4 Three-dimensional effects in the wake result in a correlation between *St* and *Re*. (**A**) Particle trace overlaying 60 images showing the recirculation region of the vortex wake ($Re = 650$). We note the three-dimensional effects in the recirculation region due to the high density of pathlines diverging from specific regions (inset). (**Bi**) We start the water tunnel after injecting dye behind the cylinder, and see the first vortex forming parallel to the cylinder as expected. (**Bii**) After the starting vortex, we see a distinct oblique shedding angle behind the cylinder form. We note that a shedding angle of approximately 10° is observed, and appears attributable to a spanwise pressure differential resulting from asymmetric boundary conditions at the ends of the cylinder. (**C**) Relationship between *St* and *Re* diverges due to oblique vortex shedding, and results in nonequivalent Strouhal periods.

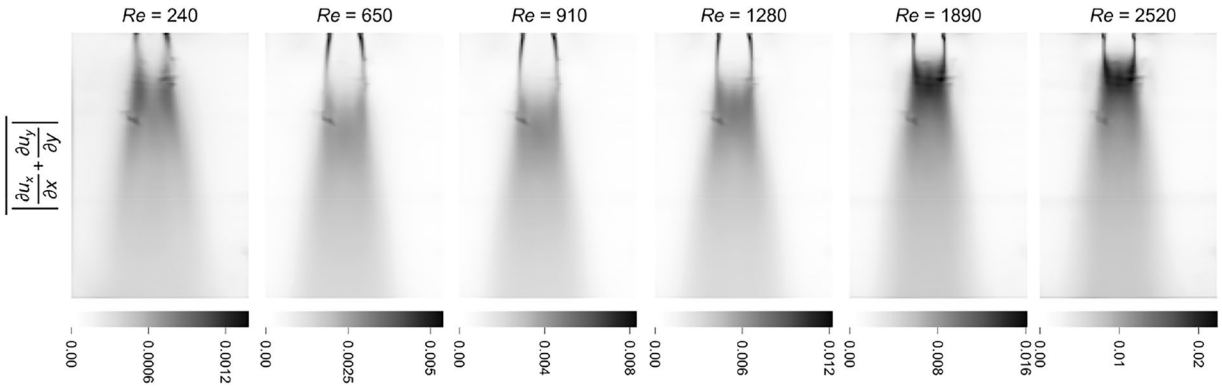


Fig. 5 Mean absolute divergence field for each Reynolds number in s^{-1}.

in length, and subsequently organized into an $n \times 100 \times 89 \times 133$ array, where each $i \times 100 \times 89 \times 133$ element represents one 100 step time sequence of 89×133 pixel (spanwise $\times$ streamwise) images. Information on the number of sequences and total amount of data for each Reynolds number is provided in Table 2.

Technical Validation

Identification of recording artefacts. The data were parsed into 100 timestep sequences to standardize the dataset and facilitate the exclusion of recording artefacts. During data collection, individual frames were occasionally corrupted or lost (<0.025% of frames); sequences are chosen so as to not include these erroneous measurements. As a result, while many consecutive sequences were recorded consecutively, there are also cases where consecutive sequences are separated by segments including dropped frames. Thus, the data cannot be restructured as a single, continuous recording of $100 \times n$ frames. Similarly, while it is often true that the ith sequence follows continuously from the $i\text{-}1^{\text{th}}$ sequence, this cannot be assumed. Artefacts were identified as frames where the mean spanwise velocity of the frame was greater than $c = 4.5$ standard deviations from the mean spanwise velocity of the entire recording:

$$|\mu_{\perp,\text{recording}} - \mu_{\perp,\text{frame}}| > c\sigma_{\perp,\text{recording}}$$

The threshold $c = 4.5$ was chosen iteratively to properly identify outlier frames. Example traces of the mean spanwise and crosswise velocities with recording artefacts identified is shown in Fig. 3.

The continuity of the sequences was further validated by checking the magnitude of the difference between consecutive velocity fields. Here, we identified and manually removed 26 additional sequences where the frames appeared out of order.

Three-dimensional effects in wake. While measured data shows only in-plane velocities, vortex shedding is known to be three-dimensional in the subcritical regime[1]. Indeed, visual inspection of the recirculation region indicates three-dimensional effects (Fig. 4A). If the flow were truly two-dimensional, there should be no points in the wake from which pathlines diverge or to which pathlines converge. Instead, we observe an area with a high density of diverging pathlines in the recirculation region, indicating particles moving into the imaging plane. Indeed, we find that the mean absolute divergence field shows increases in the shear layer and the recirculation region (Fig. 5).

Additionally, an oblique vortex shedding angle was observed when collecting data (Fig. 4B); this is known to have implications on the *St-Re* relationship[11,12]. Thus, we cannot guarantee that matching the non-dimensional time (t^*) ensures equivalent Strouhal periods (Fig. 4C, Table 1). Given these variations in Strouhal number, each 100-frame sequence represents approximately 2.5 to 3.5 vortex shedding cycles (Table 1).

Since the cylinder was fixed rigidly to the tunnel and not free to vibrate or oscillate, oblique shedding was likely the result of a pressure gradient forming from minor differences in boundary conditions at the spanwise ends of cylinder[11]. We note that there was a minor asymmetry in cylinder geometries at the two walls, where a screw was used to adjust the exact cylinder length on one side. It is also possible that the difference in boundary conditions was the result of some spanwise deviation in the incoming velocity due to tunnel geometry (e.g., a blockage or crooked flow manipulator) or pump malfunction (e.g., one pump was less efficient than another). Regardless of the exact source of the oblique shedding angle, this effect should not impact the validity of this data for training and predicting cylinder wake velocity fields. In fact, the additional three-dimensional effects may provide more challenging dynamics. We simply note that the deviation in shedding frequency was observed, and that assumptions regarding equivalency between Strouhal period and formation time are not valid across the full range of Reynolds numbers tested.

Usage Notes

A Python script is provided alongside the dataset which loads the dataset, reports the Strouhal and Reynolds numbers of each experimental condition, and generates visualizations of the velocity fields and *St-Re* relationship.

Data availability

The experimental dataset is openly available on the CaltechDATA repository (https://doi.org/10.22002/g2z1f-nex65).

Code availability

Custom code used to access data and generate figures is available for download with the dataset on the CaltechDATA repository (https://doi.org/10.22002/g2z1f-nex65). Code is available as Python scripts and was run as provided with Python 3.10.

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Author contributions

P.R. designed and conducted all experiments and processed data. P.R. and E.P. analyzed data and wrote the manuscript. P.R. and M.G. acquired funding. M.G. supervised the work.

Competing interests

The authors declare no competing interests.

Additional information

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